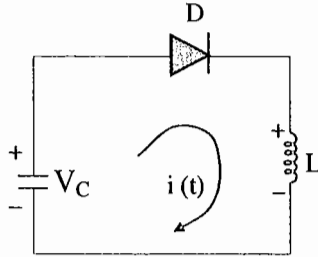


Theory of Tank circuit:



According to KVL:

$$L \frac{di(t)}{dt} + \frac{1}{C} \int i(t) dt = 0$$

Applying Laplace transform

$$s LI(s) + \frac{1}{C} \left[\frac{I(s)}{s} - \frac{CV_0}{s} \right] = 0$$

Here negative sign before $\frac{CV_0}{s}$ is used because current $i(t)$ leaves the positive terminal of C and enters its negative terminal.

$$I(s) \left[sL + \frac{1}{sC} \right] = \frac{V_0}{s}$$

$$I(s) = \frac{V_0}{L} \left[\frac{1}{s^2 + \omega_0^2} \right]$$

$$I(s) = \frac{V_0}{\omega_0 L} \cdot \left[\frac{\omega_0}{s^2 + \omega_0^2} \right]; \text{ where } \omega_0^2 = \frac{1}{LC}$$

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

By applying inverse Laplace to the above expression,

$$i(t) = \frac{V_0}{\omega_0 L} \sin \omega_0 t = \frac{V_0}{\frac{1}{\sqrt{LC}} \cdot L} \sin \omega_0 t$$

$$i(t) = V_0 \sqrt{\frac{C}{L}} \sin \omega_0 t$$

$$V_L = L \frac{di(t)}{dt} = L \frac{d}{dt} \left(\frac{V_0}{\omega_0 L} \sin \omega_0 t \right)$$

$$= \frac{L V_0}{\omega_0 L} \omega_0 \cos \omega_0 t$$

$$V_L = V_0 \cos \omega_0 t \left[\because \frac{d}{dx} \sin n\theta = n \cos n\theta \right]$$

$$V_C = \frac{1}{C} \int i(t) dt - V_0 = \frac{1}{C} \int_0^t \frac{V_0}{\omega_0 L} \sin \omega_0 t dt - V_0$$

$$= \frac{1}{C} \frac{V_0}{\omega_0 L} \left[\frac{-\cos \omega_0 t}{\omega_0} \right]_0^t - V_0$$

$$= \frac{1}{C} \frac{V_0}{\omega_0 L} \left[\frac{-\cos \omega_0 t}{\omega_0} + \frac{\cos 0}{\omega_0} \right] - V_0$$

$$= V_0 (1 - \cos \omega_0 t) - V_0$$

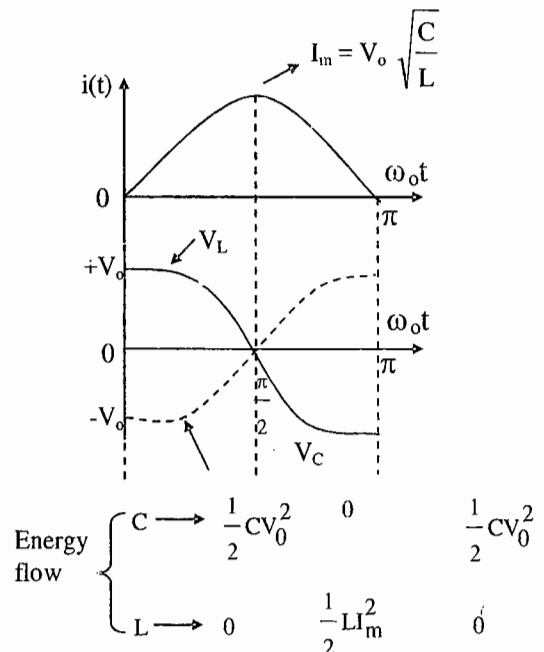
$$V_C = -V_0 \cos \omega_0 t$$

- The circuit will be conducted for half cycle i.e., π radians.

$$\omega_0 t = \pi$$

$$t = \frac{\pi}{\omega_0} = \frac{\pi}{1/\sqrt{LC}}$$

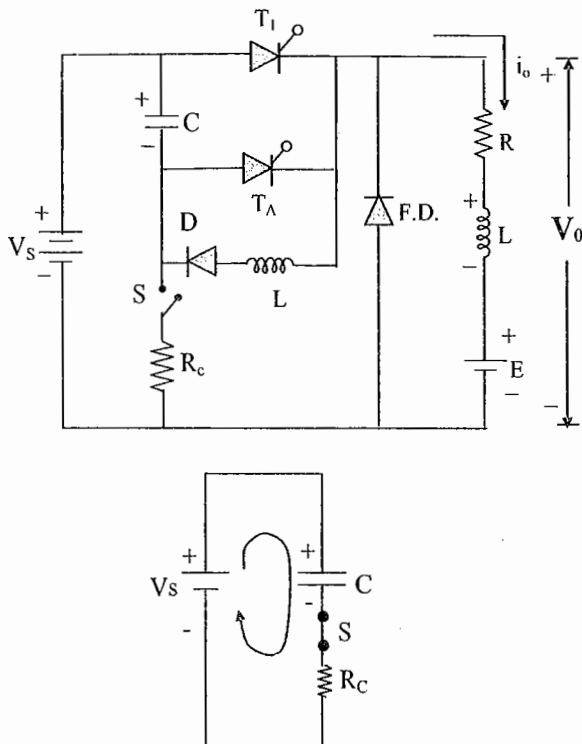
$$t = \pi \sqrt{LC} \text{ (sec)}$$



- When ever tank circuit forms, the following points are applicable
 - Nature of current is sinusoidal.
 - peak value of current is $V_0 \sqrt{C/L}$
 - conduction time of the circuit is $\pi \sqrt{LC}$ sec.
 - At the end of conduction voltage across capacitor gets reversed.

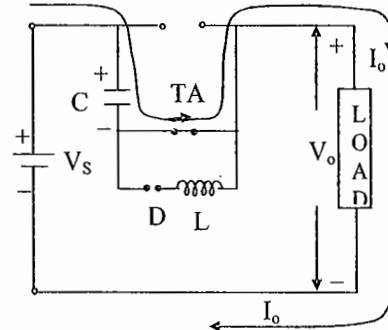
Voltage Commutated Chopper:

Circuit diagram:



- Similar to step down chopper.
 T_1 = Main Thyristor. T_A - Auxiliary thyristor.
- L, C = Commutating components, R_c - charging resistor.
- Assume output current is constant.
- Close the switch, initially capacitor short circuited, after 4-5 time constants, $V_c = V_s$.
- At $t = 0$, T_1 is on, Load is connected across the supply $V_o = V_s$.
- Tank circuit starts conduction, (Diode forward biases).
- After conduction polarities across capacitor are changed.
- D is reverse biased polarities across capacitor are changed.
- upto t_2 we completed now we have to turn off the main thyristor.
- Make T_A (Auxiliary thyristor) ON, T_1 to be OFF (applying reverse voltage).

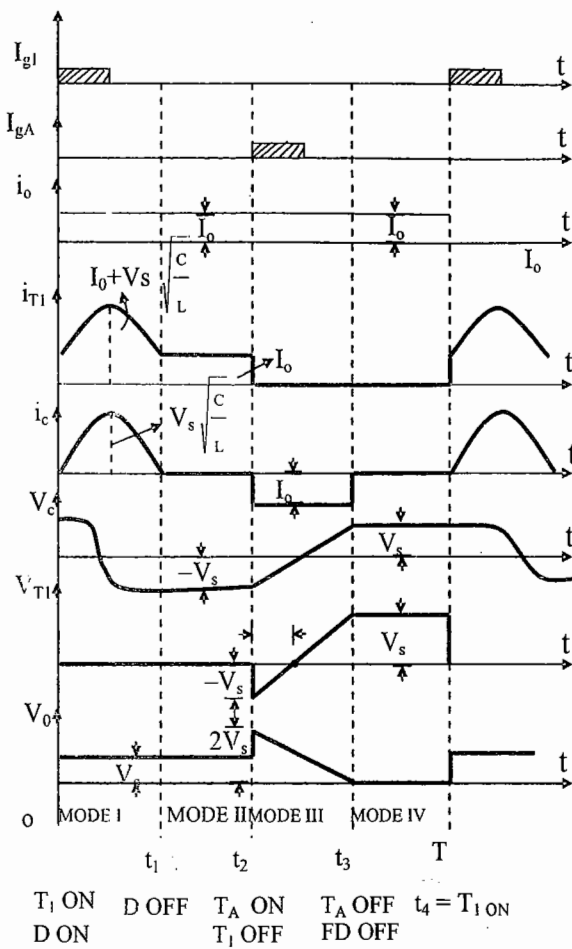
- To make the conduction continuous use free wheeling diode.
- In order to make the o/p continuous, the existing path will be changed as V_s , C , T_A and the load.



- Voltage across the capacitor changes again.
 V_s^- to V_s^+ .
- Now make the voltage across capacitor (little over charged.) $> V_s$.
- Free wheeling diode conducts, output voltage becomes zero.
- To start Next cycle, no need to close switch 'S'.
- A Reverse voltage is applied across conducting SCR due to which current through SCR becomes zero and it is getting OFF. Hence it is called voltage commutation.
- Other name of this is impulse commutation. It is because a high reverse voltage will turn off the SCR.

Limitations:

- (i) A starting circuit is required.
- (ii) Load voltage at once rises to $2V_s$ at the instant commutation of main SCR is initiated.
- (iii) It can't work at no load. It is because at no load, capacitor would not get charged from $-V_s$ to V_s when auxiliary SCR is triggered for commutating the main SCR.
- (iv) Main thyristor is required to carry current more than load current. (Because of i_c). so, it is to be over rated.
- (v) Minimum turn on time of chopper = $\pi\sqrt{LC}$.



$$V_n = \left\{ \frac{2V_s}{n\pi} \sin(n\pi\alpha) \right\} \sin(n\omega t - \theta_n)$$

- The values of commutating components C and L can be obtained :
- The values depend up on turn off time of main thyristor T_1 .
- During t_c capacitor voltage changes from $-V_s$ to zero linearly.

$$i_c = C \frac{dV}{dt}$$

for a constant Load current I_o .

$$I_o = C \frac{V_s}{t_c} \Rightarrow C = \frac{I_o t_c}{V_s}$$

- The commutation circuit turn off time ' t_c ' must be greater than thyristor turn off time.

$$t_c = t_q + \Delta t$$

$$C = \frac{(t_q + \Delta t) I_o}{V_s}$$

Commutating inductor L:

- Its value will be calculated based on peak current through main SCR.
- Peak current through main SCR = $I_o + V_s \sqrt{\frac{C}{L}}$
- Hence load current should not be too large.

$$I_{cp} \leq I_o ; V_s \sqrt{\frac{C}{L}} \leq I_o ; \frac{V_s \sqrt{C}}{I_o} \leq \sqrt{L}$$

$$L \geq \left(\frac{V_s}{I_o} \right)^2 C$$

- peak current through Auxiliary thyristor = I_o
- peak current through Diode = $V_s \sqrt{\frac{C}{L}}$
- peak current through capacitor = $V_s \sqrt{\frac{C}{L}}$
- peak instantaneous output voltage = $2V_s$.

PIV Rating of FW Diode = $2V_s$.

Average load voltage V_o

$$= \left[\frac{V_s T_{on} + \left(\frac{t_3 - t_2}{2} \right) 2V_s}{T} \right]$$

$$= \frac{V_s}{T} [T_{on} + (t_3 - t_2)]$$

During the time interval $t_3 - t_2$ voltage across 'C' changes from $-V_s$ to V_s i.e. total change is $2V_s$.

$$I_o = C \frac{2V_s}{(t_3 - t_2)} (t_3 - t_2) = C \frac{2V_s}{I_o}$$

$$V_o = \frac{V_s}{T} \left[T_{on} + C \frac{2V_s}{I_o} \right]$$

$$V_o = V_s \left(\frac{T_{on}'}{T} \right) = \alpha' V_s$$

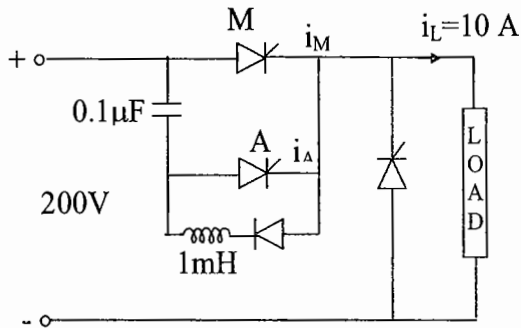
Where T_{on}' = Effective turn ON time

$$T_{ON}' = T_{ON} + \frac{2CV_s}{I_o}$$

Where α' = Effective duty cycle

Workedout Example

01. A voltage commutated chopper circuit, operated at 500HZ, is shown below. If the maximum value of load current is 10A, then the maximum current through the main (M) and auxiliary (A) thyristor will be



- (a) 12A, 10A (b) 12A, 2A
(c) 10A, 12A (d) 10A, 8A

Sol: Current through main thyristor is $\frac{V_s}{R} + V_s \sqrt{\frac{C}{L}}$

$$= 10 + 200 \sqrt{\frac{0.1 \times 10^{-6}}{1 \times 10^{-3}}}$$

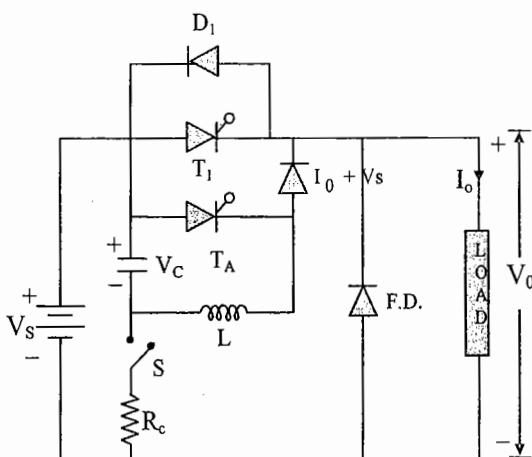
$$= 12 \text{ A}$$

Current through auxiliary thyristor is $V_s \sqrt{\frac{C}{L}}$

$$= 200 \sqrt{\frac{0.1 \times 10^{-6}}{1 \times 10^{-3}}}$$

$$= 10 \text{ A}$$

Current Commutated Chopper:



- Capacitor is charged to V_s , main thyristor T_1 is fired at $t = 0$. So that load voltage $V_o = V_s$
- At $t = t_1$, auxiliary thyristor is turned on to commute main thyristor.
- with turning on of T_A , an Oscillatory Current (i_c) is set up in the circuit.

$$i_c = \frac{V_s}{\omega_o L} \sin \omega_o t = V_s \sqrt{\frac{C}{L}} \sin \omega_o t.$$

- At t_2 , $V_c = -V_s$ and i_c tends to reverse in the auxiliary thyristor T_A , it gets naturally commutated.
- As T_A is reverse biased and turned off at t_2 . Oscillatory current i_c begins to flow through C , L , D_2 and T_1 .
- At t_3 i_c rises to I_o so that $i_{T1} = 0$. As a result main SCR T_1 is turned off at t_3 .
- Since Oscillating current through T_1 turns it off it is called current commutated chopper.
- After t_3 i_c supplies load current I_o and the excess current.
- $i_{D1} = i_c - I_o$ is conducted through diode D_1
- After t_4 , a constant current equal to I_o flows through V_s , C , L , D_2 and load.
- capacitor C is charged linearly to. Source voltage V_s at t_5 , so during time $(t_5 - t_4)$

$$i_c = I_o.$$

- In this commutation an opposite current pulse will be injected through SCR. As a result currents decreases and finally comes to zero if both the currents would be equal and opposite.
- Anti parallel diode is useful to apply the reverse voltage after current through SCR becomes to zero. The value of reverse voltage is low. So

(1) Turn off time increases.

(2) Turn off power loss increases.

$$L = \frac{V_s \cdot t_c}{(x I_o) \left[\pi - 2 \sin^{-1} \left(\frac{1}{x} \right) \right]}$$

where $x = \frac{I_{cp}}{I_0}$ and

$$C = \frac{(x I_o) t_c}{V_s \left[\pi - 2 \sin^{-1} \left(\frac{1}{x} \right) \right]}$$

- '*Jones chopper*' employs the principle of voltage commutation.
- Morgan's chopper based on the principle of current commutation.

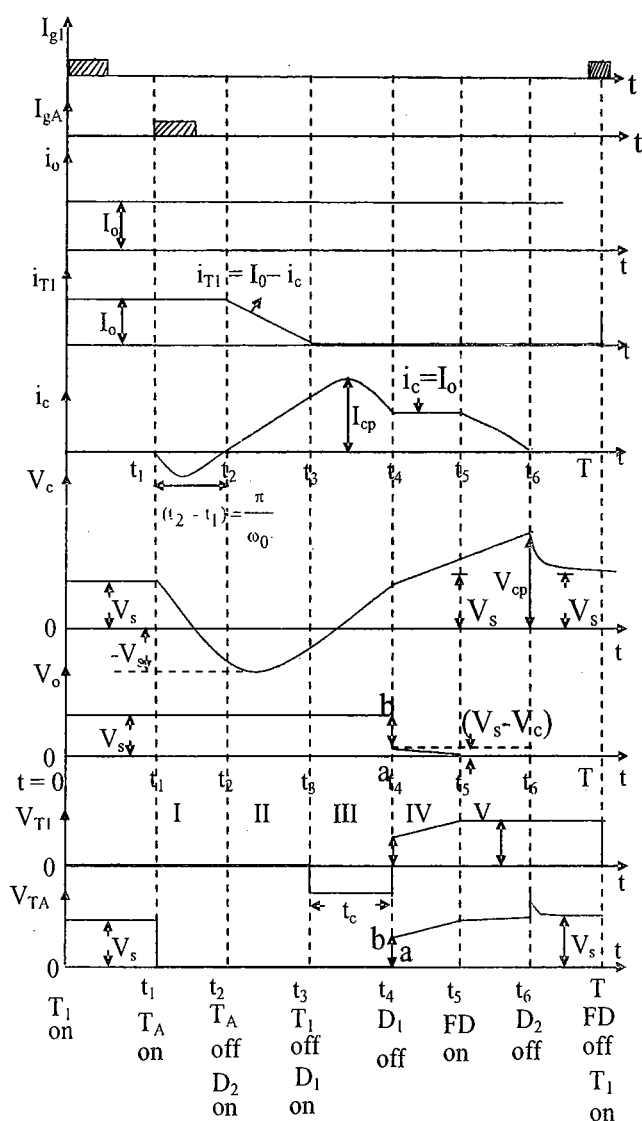


Fig. Current and voltage waveforms for current commutated chopper

Workedout Examples

02. In a resonant commutation circuit, supply voltage is 200V. Load current is 10A and the device turn off time is 30 μ s. The ratio of peak resonant current to load current is 1.5. Find the value of L&C

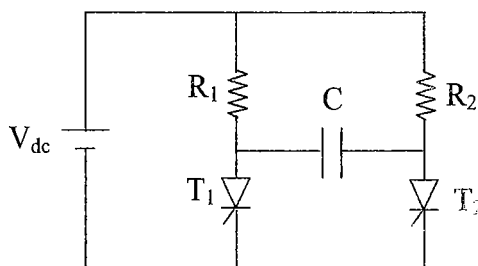
- (a) $L = 2.66 \text{ mH}$ & $C = 1.5 \text{ }\mu\text{F}$
 (b) $L = 1.5 \text{ mH}$ & $C = 0.266 \text{ }\mu\text{F}$
 (c) $L = 0.266 \text{ mH}$ & $C = 1.5 \text{ }\mu\text{F}$
 (d) None of these

Sol: $I_p = 1.5I_0 = 15A = V_{dc} \sqrt{\frac{C}{L}}$

$$t_c = \frac{C \cdot V_{dc}}{I_0} \Rightarrow 30 \mu = \frac{C \times 200}{10} \Rightarrow C = 1.5 \mu F$$

$$\therefore L = 0.266 \text{ mH}$$

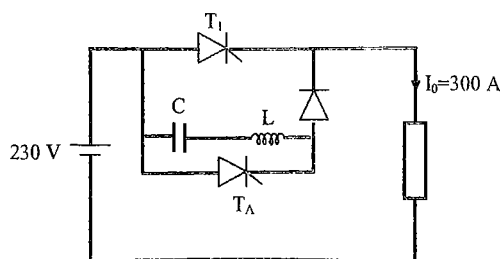
03. In the following commutation circuit, $R_1 = R_2 = R = 5 \Omega$ and the capacitance $C = 7.5 \mu F$, $V_{dc} = 100 V$. Determine the circuit turn off time.



Sol: $R_1 = R_2 = R = 5\Omega$

$$\begin{aligned}\text{Circuit turn off time} &= R_C \log 2 \\ &= 5 \times 7.5 \times 10^{-6} \log 2 \\ &= 26 \mu \text{ sec}\end{aligned}$$

04. In the following resonant pulse communication scheme, Find the voltage across main SCR when it gets turned off.



$$L=5 \mu H \quad \& \quad C=20 \mu F$$



- (a) 460 V (b) 177 V
(c) 174.37 V (d) 230 V

Sol: Peak value of resonant current

$$I_P = V_s \sqrt{\frac{C}{L}} = 230 \sqrt{\frac{20}{5}} = 460 \text{ A}$$

$$\begin{aligned} \text{Resonant frequency } \omega_0 &= \frac{1}{\sqrt{LC}} \\ &= \frac{1}{\sqrt{20 \times 10^{-6} \times 5 \times 10^{-6}}} \\ &= 0.1 \times 10^6 \text{ rad/sec} \end{aligned}$$

$$\sin^{-1} \left(\frac{I_0}{I_P} \right) = \omega_0 (t_3 - t_2)$$

$$\sin^{-1} \left(\frac{300}{460} \right) = 0.1 \times 10^{-6} (t_3 - t_2)$$

$$t_3 - t_2 = 40.706^\circ$$

Voltage across main thyristor, when it gets turned off is

$$\begin{aligned} V_{ab} &= V_s \cos \omega_0 (t_3 - t_2) \\ &= 230 \cos(40.706^\circ) \\ &= 174.355 \text{ V} \end{aligned}$$